

CS 660: Data Mining For Decision Making

Lecture 6-7 (Week 3)

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Scaling the Heights

Decision Tree Classifiers: An Alternative Rational Approach

- Often times, we want to use production rules (if – then rules for classification) for making decisions (In decision-making literature they are also referred to as “heuristics”)
- Suppose, we have a set of attributes characterizing a stimulus (E.g., Professor or Student). The idea is to build a decision tree and then use this tree to generate the production rules (if – then rules for classification)
- There are several ways of building decision trees based upon the order in which the attributes are considered
- However, based upon simple mathematics, one could find out the best order of considering attributes in a decision tree and thus figure out the best possible production rules from a training dataset

Decision Task: Classification as Student or Professor (Training Set)

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	Clogs	Student
T6	Quiet	Baked	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

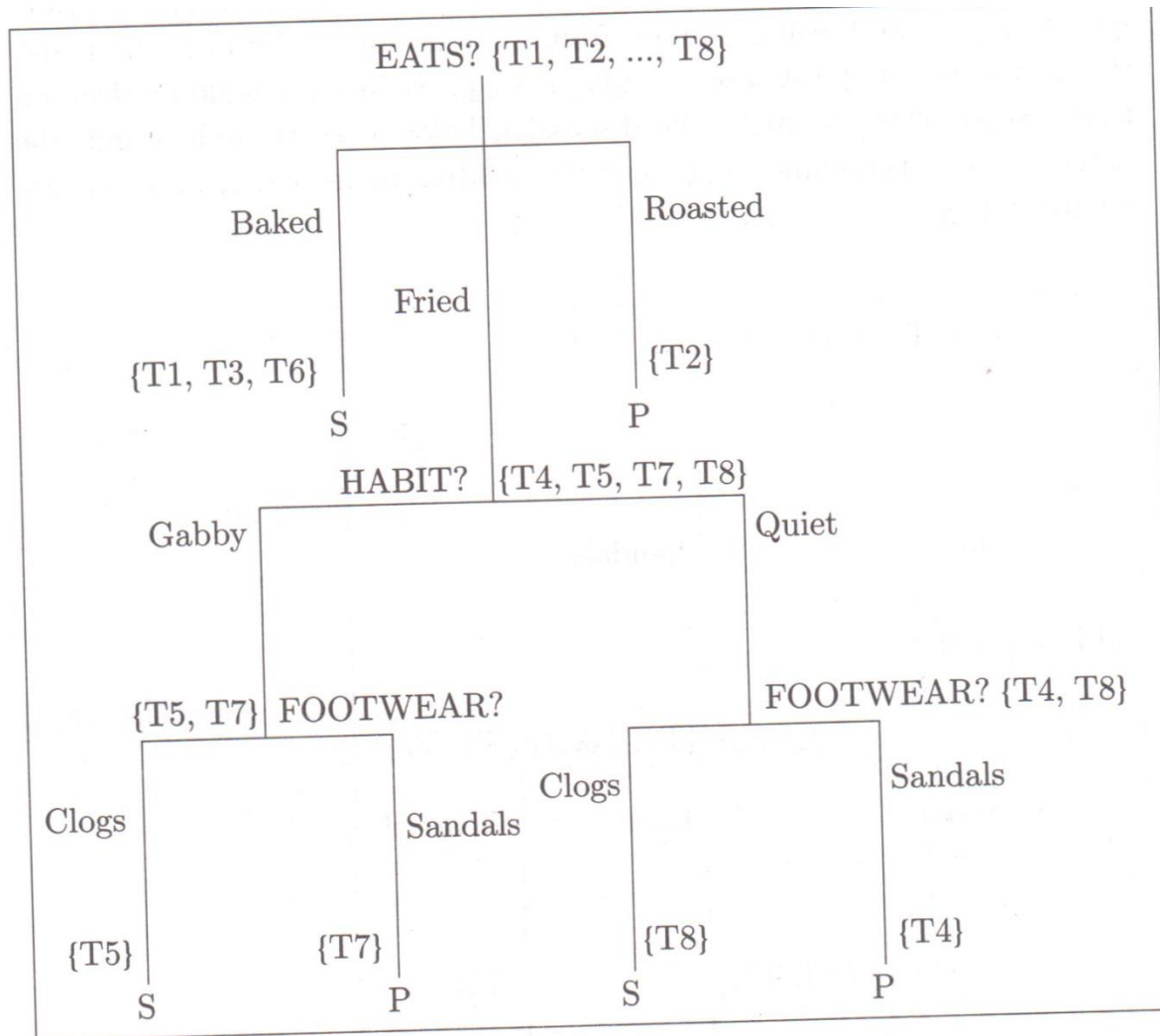
FIGURE 1.1 Training set for recognizing professors and students. The possible attribute values are 'gabby' and 'quiet' for HABIT; 'baked', 'fried', and 'roasted' for EATS; and 'clogs' and 'sandals' for FOOTWEAR.

Decision Task: Classification as Student or Professor (Test/Generalization Set)

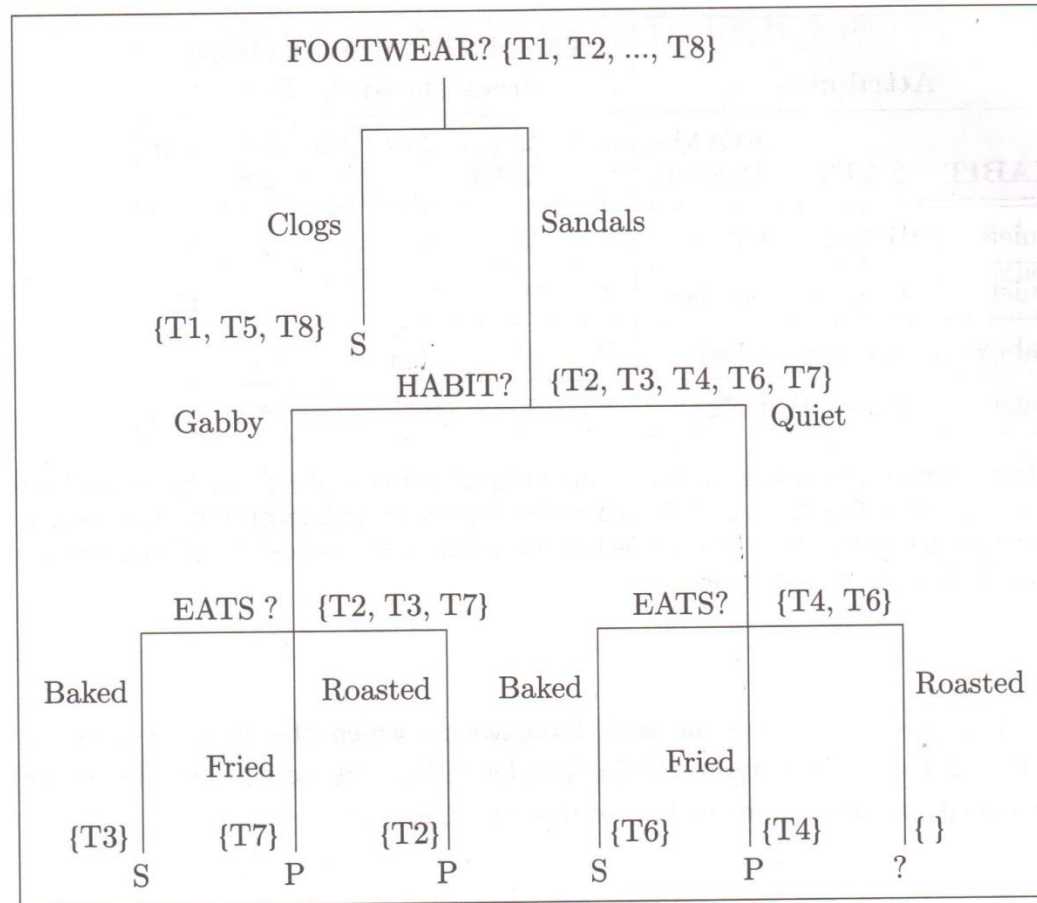
NAME of recall pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	
R2	Quiet	Roasted	Sandals	
R3	Gabby	Roasted	Clogs	
R4	Quiet	Roasted	Clogs	

FIGURE 1.2 Recall set for recognizing professors and students. The training set is given in Fig. 1.1.

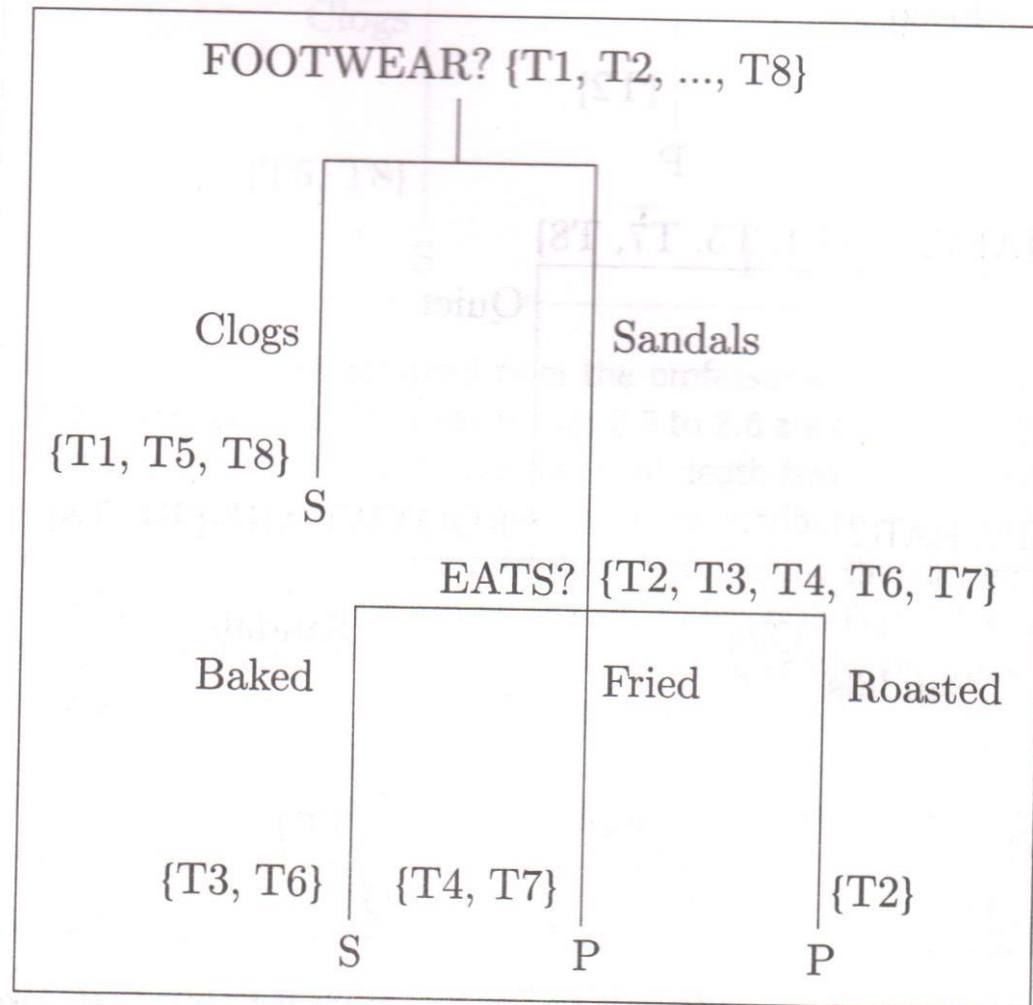
A Decision Tree



Another Decision Tree



Still Another Decision Tree



How to decide the best sequence of attributes in a decision tree?

$$\text{Maximize the Ratio of Information Gain} = (G_A(V_0)) = \frac{I(V_0) - I_A(V_0)}{S_A(V_0)}$$

So, the average information needed by attribute A to split set V_0 into sets V_1 to V_n is

$$S_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} \quad (2.4)$$

The average information required to classify a pattern in set V_0 into one of the m classes is then expressed as

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} \quad (2.1)$$

Let $C_1, C_2, \dots, C_m$ be the $m \geq 1$ classes to which the training patterns belong. For $1 \leq k \leq m$ and $0 \leq i \leq n$, let $Y(k, i)$ be the number of training patterns of class C_k in set V_i . For $0 \leq i \leq n$, let

$$I_A(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} \quad (2.2)$$

$$Z(i) = \sum_{k=1}^m Y(k, i)$$

Calculating $I(V_0)$

To evaluate the entropy $I(V_0)$ of the professor-student training set of Fig. 1.1. The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Three patterns belong to C_1 and five to C_2 . Therefore,

The training set $V_0 = \{T1, T2, \dots, T8\}$

$Y(1,0) = 3$ (number of patterns in V_0 of class C_1)

$Y(2,0) = 5$ (number of patterns in V_0 of class C_2)

$Z(0) = 8$ (number of patterns in V_0)

From Eqn (2.1) in Section 2.5, the average information required to classify a pattern in V_0 into one of the m classes is

$$\begin{aligned} I(V_0) &= - \sum_{k=1}^m \frac{Y(k,0)}{Z(0)} \log \frac{Y(k,0)}{Z(0)} \\ &= - \sum_{k=1}^2 \frac{Y(k,0)}{Z(0)} \log \frac{Y(k,0)}{Z(0)} \\ &= -\frac{3}{8} \log \frac{3}{8} - \frac{5}{8} \log \frac{5}{8} \\ &= 0.375 \times 1.415 + 0.625 \times 0.6780 \\ &= 0.5306 + 0.4238 \\ &= 0.9544 \end{aligned}$$

Splitting on Habit Attribute

To evaluate $G_{\text{HABIT}}(V_0)$ at the root of a decision tree for the professor-student training set of Fig. 1.1, the following procedure is used. The root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since HABIT has two possible values (gabby and quiet), root y_0 has two children y_1 and y_2 . Therefore, $n = 2$. To understand the following material easily, one can look at any one of the decision trees shown in Figs 2.3 to 2.6, since each of them examines HABIT at the root.

$V_0 = \{T1, T2, \dots, T8\}$ at node y_0

$Y(1, 0) = 3$ (number of patterns in V_0 of class C_1)

$Y(2, 0) = 5$ (number of patterns in V_0 of class C_2)

$Z(0) = 8$ (number of patterns in V_0)

$V_1 = \{T1, T2, T3, T5, T7\}$ at node y_1 , where HABIT = gabby

$Y(1, 1) = 2$ (number of patterns in V_1 of class C_1)

$Y(2, 1) = 3$ (number of patterns in V_1 of class C_2)

$Z(1) = 5$ (number of patterns in V_1)

$V_2 = \{T4, T6, T8\}$ at node y_2 , where HABIT = quiet

$Y(1, 2) = 1$ (number of patterns in V_2 of class C_1)

$Y(2, 2) = 2$ (number of patterns in V_2 of class C_2)

$Z(2) = 3$ (number of patterns in V_2)

$I(V_0) = 0.9544$ (from Fig. 2.12)

$$I_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.9499 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{HABIT}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{HABIT}}(V_0) = \frac{I(V_0) - I_{\text{HABIT}}(V_0)}{S_{\text{HABIT}}(V_0)} = 0.0047 \quad [\text{from Eqn (2.5)}]$$

Splitting on Eats Attribute

To evaluate $G_{\text{EATS}}(V_0)$ at the root of a decision tree for the professor-student training set, the following procedure is used. The root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since EATS has three possible values (baked, fried, and roasted), root y_0 has three children y_1 , y_2 , and y_3 . Therefore, $n = 3$. To understand the following material easily, one can look at either of the decision trees shown in Figs 2.7 or 2.8, since each of them examines EATS at the root.

$V_0 = \{T1, T2, \dots, T8\}$ at node y_0

$Y(1, 0) = 3$ (number of patterns in V_0 of class C_1)

$Y(2, 0) = 5$ (number of patterns in V_0 of class C_2)

$Z(0) = 8$ (number of patterns in V_0)

$V_1 = \{T1, T3, T6\}$ at node y_1 , where EATS = baked

$Y(1, 1) = 0$ (number of patterns in V_1 of class C_1)

$Y(2, 1) = 3$ (number of patterns in V_1 of class C_2)

$Z(1) = 3$ (number of patterns in V_1)

$V_2 = \{T4, T5, T7, T8\}$ at node y_2 , where EATS = fried

$Y(1, 2) = 2$ (number of patterns in V_2 of class C_1)

$Y(2, 2) = 2$ (number of patterns in V_2 of class C_2)

$Z(2) = 4$ (number of patterns in V_2)

$V_3 = \{T2\}$ at node y_3 , where EATS = roasted

$Y(1, 3) = 1$ (number of patterns in V_3 of class C_1)

$Y(2, 3) = 0$ (number of patterns in V_3 of class C_2)

$Z(3) = 1$ (number of patterns in V_3)

$I(V_0) = 0.9544$ (from Fig. 2.12)

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.5 \text{ [from Eqn (2.2)]}$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.4055 \text{ [from Eqn (2.4)]}$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.3233 \text{ [from Eqn (2.5)]}$$

Splitting on Footwear Attribute

To evaluate $G_{\text{FOOTWEAR}}(V_0)$ at the root of a decision tree for the professor-student training set, the following procedure is used. The root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since FOOTWEAR has two possible values (clogs and sandals), root y_0 has two children y_1 and y_2 . Therefore, $n = 2$. To understand the following material easily, one can look at either of the decision trees shown in Fig. 2.9 or 2.10, since each of them examines FOOTWEAR at the root.

$V_0 = \{T1, T2, \dots, T8\}$ at node y_0

$Y(1, 0) = 3$ (number of patterns in V_0 of class C_1)

$Y(2, 0) = 5$ (number of patterns in V_0 of class C_2)

$Z(0) = 8$ (number of patterns in V_0)

$V_1 = \{T1, T5, T8\}$ at node y_1 , where FOOTWEAR = clogs

$Y(1, 1) = 0$ (number of patterns in V_1 of class C_1)

$Y(2, 1) = 3$ (number of patterns in V_1 of class C_2)

$Z(1) = 3$ (number of patterns in V_1)

$V_2 = \{T2, T3, T4, T6, T7\}$ at node y_2 , where FOOTWEAR = sandals

$Y(1, 2) = 3$ (number of patterns in V_2 of class C_1)

$Y(2, 2) = 2$ (number of patterns in V_2 of class C_2)

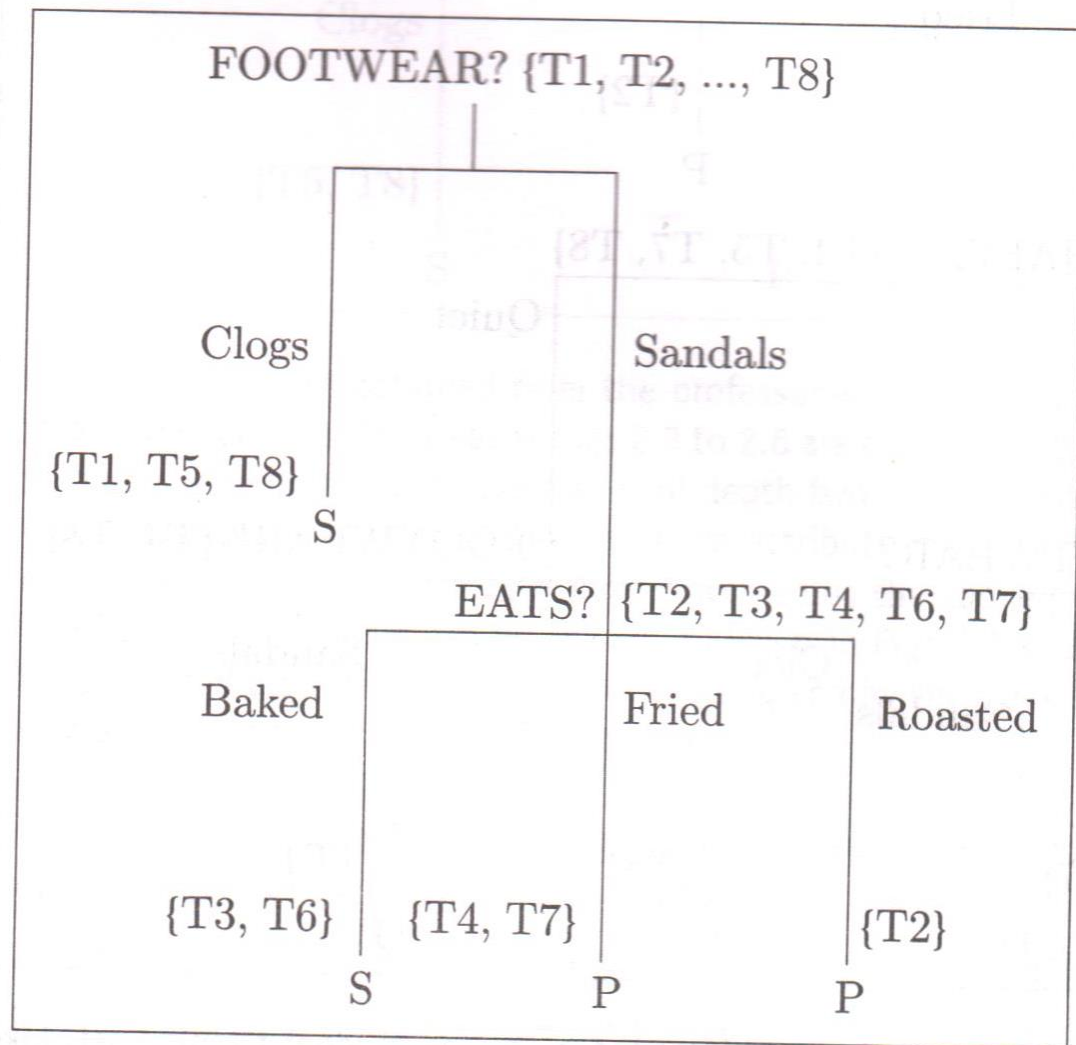
$Z(2) = 5$ (number of patterns in V_2)

$I(V_0) = 0.9544$ (from Fig. 2.12)

$$I_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0.6066 \text{ [from Eqn (2.2)]}$$

$$S_{\text{FOOTWEAR}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 0.9544 \text{ [from Eqn (2.4)]}$$

$$G_{\text{FOOTWEAR}}(V_0) = \frac{I(V_0) - I_{\text{FOOTWEAR}}(V_0)}{S_{\text{FOOTWEAR}}(V_0)} = 0.3640 \text{ [from Eqn (2.5)]}$$



Splitting on Eats Attribute At Next Level

To evaluate $G_{\text{EATS}}(V_0)$ at the right child of the root of a decision tree for the professor-student training set, the following procedure is used. FOOTWEAR has already been examined at the root (see Fig. 2.9). The right child of the root is named y_0 . The classes are C_1 (professor) and C_2 (student). Therefore, $m = 2$. Since EATS has three possible values (baked, fried, and roasted), node y_0 has three children y_1 , y_2 , and y_3 . Therefore, $n = 3$.

$V_0 = \{T2, T3, T4, T6, T7\}$ at root y_0
 $Y(1, 0) = 3$ (number of patterns in V_0 of class C_1)
 $Y(2, 0) = 2$ (number of patterns in V_0 of class C_2)
 $Z(0) = 5$ (number of patterns in V_0)
 $V_1 = \{T3, T6\}$ at node y_1 , where EATS = baked
 $Y(1, 1) = 0$ (number of patterns in V_1 of class C_1)
 $Y(2, 1) = 2$ (number of patterns in V_1 of class C_2)
 $Z(1) = 2$ (number of patterns in V_1)
 $V_2 = \{T4, T7\}$ at node y_2 , where EATS = fried
 $Y(1, 2) = 2$ (number of patterns in V_2 of class C_1)
 $Y(2, 2) = 0$ (number of patterns in V_2 of class C_2)
 $Z(2) = 2$ (number of patterns in V_2)
 $V_3 = \{T2\}$ at node y_3 , where EATS = roasted
 $Y(1, 3) = 1$ (number of patterns in V_3 of class C_1)
 $Y(2, 3) = 0$ (number of patterns in V_3 of class C_2)
 $Z(3) = 1$ (number of patterns in V_3)

$$I(V_0) = - \sum_{k=1}^m \frac{Y(k, 0)}{Z(0)} \log \frac{Y(k, 0)}{Z(0)} = 0.9710 \quad [\text{from Eqn (2.1)}]$$

$$I_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \sum_{k=1}^m \frac{Y(k, j)}{Z(j)} \log \frac{Y(k, j)}{Z(j)} = 0 \quad [\text{from Eqn (2.2)}]$$

$$S_{\text{EATS}}(V_0) = - \sum_{j=1}^n \frac{Z(j)}{Z(0)} \log \frac{Z(j)}{Z(0)} = 1.5218 \quad [\text{from Eqn (2.4)}]$$

$$G_{\text{EATS}}(V_0) = \frac{I(V_0) - I_{\text{EATS}}(V_0)}{S_{\text{EATS}}(V_0)} = 0.6377 \quad [\text{from Eqn (2.5)}]$$

Optimal Production Rules

If FOOTWEAR = clogs,
then the pattern belongs to the class student.

If FOOTWEAR = sandals,
and EATS = baked,
then the pattern belongs to the class student.

If FOOTWEAR = sandals,
and EATS = fried,
then the pattern belongs to the class professor.

If FOOTWEAR = sandals,
and EATS = roasted,
then the pattern belongs to the class professor.



Questions or Comments?